

Construction of a Joint Simulation Platform Using SUMO, CARLA, and Autoware and Its Application in Multi-Vehicle Interactions at Unsignalized Intersections

Bo Liu

School of Automotive and
Mechatronic Engineering
Harbin Cambridge University
Harbin, Heilongjiang, China
bliu33@gmail.com

Jian He

School of Intelligent and
Electrical-Electronic Engineering
Harbin Cambridge University
Harbin, Heilongjiang, China

Hao Li

Company of Qingfei.AI
Hangzhou, Zhejiang, China
Corresponding author: haoliarts@gmail.com

Haoyang Duan

School of Automotive and
Mechatronic Engineering
Harbin Cambridge University
Harbin, Heilongjiang, China

Rongrong Li

School of Automotive and
Mechatronic Engineering
Harbin Cambridge University
Harbin, Heilongjiang, China

Yifei Sun

Company of Qingfei.AI
Hangzhou, Zhejiang, China

Abstract—This study aims to construct a joint simulation platform based on SUMO, CARLA, and Autoware, and apply the platform to the investigation of multi-vehicle game scenarios at unsignalized intersections. By leveraging the strengths of these three open-source tools, we created an efficient environment for modeling complex traffic scenarios and evaluating autonomous driving algorithms with a low-code approach. The research primarily focuses on reproducing deadlock situations encountered by Robotaxis during public road tests and assessing the impact of vehicle-to-infrastructure (V2X) cooperation on traffic efficiency through comparative experiments. The results demonstrate that the proposed platform effectively simulates multi-vehicle interactions, serving as a valuable tool for optimizing autonomous driving systems.

Keywords—Joint Simulation; Autonomous Driving; Multi-Vehicle Interaction; SUMO; CARLA; Autoware; Unsignalized Intersection

I. INTRODUCTION

The rapid advancement of autonomous driving technology has intensified research on the safe and efficient deployment of autonomous vehicles in real-world traffic [1]. Complex unsignalized intersections often lead to low traffic efficiency or complete deadlock due to the competitive behavior of multiple vehicles. Many scholars have focused their efforts on this research direction. However, for the majority of researchers, conducting such studies in real traffic environments is impractical, let alone replicating these intersection game-theoretic traffic scenarios in real traffic settings. This study

aims to construct a joint simulation platform that combines SUMO, CARLA, and Autoware [2][3][4][5][6], thereby providing researchers with a tool to efficiently implement multi-vehicle interactions scenarios without the need to develop a comprehensive autonomous driving code.

The significance of constructing a joint simulation platform is threefold. Firstly, it allows for the simulation of intricate real-world traffic scenarios, providing a safe and controlled environment for the development and validation of autonomous driving algorithms. Secondly, by integrating various simulation tools, researchers can comprehensively evaluate the performance of autonomous systems with low-code approach. Lastly, the platform facilitates investigations into the effects of advanced technologies, such as V2X cooperation, on traffic efficiency.

II. CONSTRUCTION OF THE JOINT SIMULATION PLATFORM

A. Overview of Three Open-Source Tools

The three open-source tools utilized in this study each contribute unique capabilities, making them well-suited for constructing a comprehensive simulation platform [7][8].

SUMO (Simulation of Urban MObility) is a microscopic traffic simulation software that excels in modeling large-scale traffic flows and collecting detailed traffic data, such as average travel time and speed.

CARLA (CAR Learning to Act) is a high-fidelity autonomous driving simulation platform built on Unreal

The Scientific Research Foundation of Harbin Cambridge University (2024JQKY02)

Engine, providing realistic visual environments and physical characteristics.

Autoware is an open-source software stack for autonomous driving, encompassing modules for perception, localization, planning, and control.

B. Architecture Design of the Joint Simulation Platform

The architecture design of the joint simulation platform is crucial for conducting efficient multi-vehicle simulations, as shown in Fig. 1. The platform integrates SUMO and CARLA via the TraCI interface, then integrates CARLA and Autoware through ROS (Robot Operating System), ensuring seamless communication and data exchange. SUMO serves as the core for traffic management, constructing the basic road network and defining macroscopic traffic parameters. It generates detailed road maps, including lane numbers, speed limits, and turning rules, while allowing flexible settings for vehicle departure times and route planning.

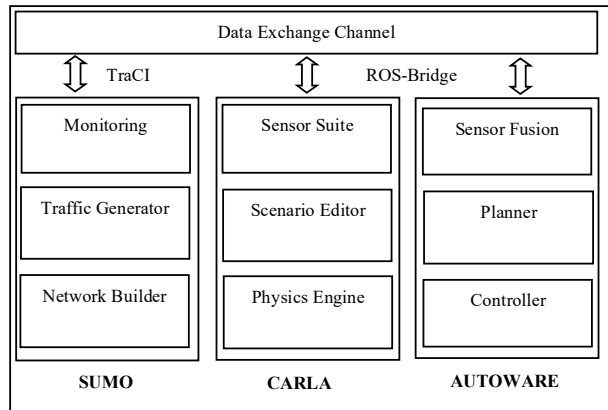


Figure 1. Architecture of Joint Simulation Platform

CARLA focuses on high-fidelity vehicle dynamics and sensor simulations. It accurately models vehicle behaviors, ensuring realistic movement in the simulation. Autoware functions as the autonomous driving algorithm platform, processing sensor data from CARLA to identify objects such as vehicles and pedestrians. Localization combines sensor data with map information to determine vehicle position. Based on this processed data, the planning module generates driving paths and speed plans, which the control module translates into vehicle control signals.

Data exchange between SUMO and CARLA occurs via the TraCI interface, ensuring real-time synchronization. SUMO sends traffic flow information to CARLA, which adjusts its simulation accordingly. Feedback regarding vehicle collisions or trajectory deviations is relayed back to SUMO to update traffic statistics. Communication between CARLA and Autoware employs ROS topics and services, allowing for efficient data processing and control signal transmission.

C. Data interaction of the Joint Simulation Platform

Data Interaction between SUMO and CARLA: SUMO and CARLA are connected via the TraCI interface, enabling real-time synchronization between the two. SUMO sends traffic flow information, such as the current IDs, positions, speeds,

directions of vehicles, and the status of traffic lights (although the signal status is fixed at non-signal intersections, this information is crucial in more complex scenarios)—packaged in a specific data format (like XML) to CARLA at predetermined time steps. Upon receiving this information, CARLA adjusts the generation and driving paths of vehicles in its simulation environment based on SUMO's traffic flow state, ensuring consistency with the macro traffic flow of SUMO. At the same time, CARLA provides real-time status updates of vehicles, such as collision detection results and whether vehicles have deviated from their preset trajectories, back to SUMO. Once SUMO receives this feedback, it can further update traffic flow statistics, providing more accurate data support for subsequent traffic analysis.

Data Interaction between CARLA and Autoware: Data exchange between the two is achieved through ROS topics and services. CARLA publishes sensor data, such as images captured by cameras and point cloud data generated by radar, to the corresponding topics according to ROS topic specifications. Autoware subscribes to these topics, processes the sensor data in its internal perception module. After Autoware processes the sensor data and generates control commands (such as acceleration, deceleration, and turning commands), it sends these commands to CARLA via ROS services. Upon receiving the control commands, CARLA converts them into actual control signals for the vehicles, enabling them to drive in the simulation environment.

Using SUMO's statistical capabilities, average speed, total travel time, and average travel time traffic efficiency metrics can be collected, providing data support for subsequent experimental analysis. During simulation, SUMO continuously records each vehicle's trajectory and time information. After the simulation ends, it calculates various traffic efficiency metrics using the recorded data through built-in algorithms. These metric data can be easily exported, facilitating further data analysis and visualization by researchers.

D. Benefits of the Joint Simulation Platform

Scholars have tried to build joint simulation platforms such as SUMO-CARLA or CARLA-Autoware [9]. However, these platforms are typically constructed in a Linux environment, which can be cumbersome for system setup and maintenance. We have successfully established a tripartite joint simulation platform integrating SUMO, CARLA, and Autoware under the Windows 11 environment, specifically utilizing the Windows Subsystem for Linux (WSL) within Windows 11 for Autoware installation. This setup provides researchers with a highly user-friendly simulation environment without the need for extensive code development. Additionally, since all three simulation tools are open-source, researchers can customize and develop their own code for more complex simulations.

Building a simulation platform in our approach offers numerous advantages. The key benefits are as follows:

1) User-Friendly Interface. This provides an intuitive user interface that simplifies the setup and maintenance of the simulation platform. Users can easily set up the simulation environment without delving into the complex configurations of the Linux system. This ease of use reduces the learning

curve, allowing researchers to focus more on the simulation tasks themselves.

2) Efficient Resource Management and Performance Optimization. The platform supports efficient resource management, enabling flexible allocation of CPU, memory, and storage resources to ensure the smooth operation of simulation tasks.

3) Powerful Graphics Processing Capabilities. It supports high-end GPU cards and professional graphics acceleration cards, which can meet the demands for complex 3D model rendering and high-resolution texture loading. This capability is particularly important for simulation tools like CARLA that have high requirements for graphics rendering.

4) Flexibility of Open-Source Software. Simulation tools such as SUMO, CARLA, and Autoware are all open-source, allowing users to customize and develop them according to their needs. In the Windows environment, the combination of this flexibility and ease of use provides researchers with powerful tool support.

In summary, the proposed approach not only provides efficient resource management, powerful graphics processing capabilities, and a user-friendly interface but also enhances system flexibility and compatibility through the WSL feature. And the whole platform can be easily migrated from one machine to another using the import and export features of WSL. Fig. 2 illustrates the visual effects of the co-simulation platform during operation. The three open-source software tools have achieved synchronization of the simulation interface.



Figure 2. Visual effect of Joint Simulation Platform

III. EXPERIMENTAL DESIGN AND METHODS

After setting up the co-simulation platform, we can begin various simulation attempts. In the environment of the proposed co-simulation platform, researchers can quickly launch a variety of experiments and verify their ideas without writing any code or with minimal coding. Next, we will apply this co-simulation platform to a multi-vehicle game scenario at an unsignalized intersection.

A. Experimental Scenario

The experimental scenario was designed to simulate a T-junction located within the CARLA Town01 map, intentionally devoid of any traffic signals to replicate real-world unsignalized intersections. This choice of setting is crucial as unsignalized intersections present unique challenges for autonomous vehicles, primarily due to the lack of explicit traffic control and the need for vehicles to negotiate right-of-way among themselves. In this scenario, vehicles approach the intersection from multiple directions, creating complex interactions influenced by factors such as vehicle speed, arrival times, and driver behavior.

To enhance the realism of the simulation, various traffic conditions were introduced, such as varying vehicle types, including passenger cars, buses, and trucks. Each vehicle type has distinct characteristics which are configured via SUMO. Additionally, the simulation included rainy environmental conditions which were set up by CARLA.

B. Experimental Groups

Two distinct experimental groups were established to evaluate the impact of V2X cooperation on traffic efficiency at the unsignalized intersection. The first group, designated as the non-cooperation group, consisted of vehicles that relied solely on their own onboard sensors and algorithms for decision-making. These vehicles were unable to communicate with each other or gain insights into the overall traffic state, which often led to inefficient responses during critical moments at the intersection. For example, when multiple vehicles approached the intersection simultaneously, they frequently encountered deadlock situations, severely hampering traffic flow.

Conversely, the second group, referred to as the cooperation group, utilized V2X communication technology, enabling vehicles to share information regarding their positions, speeds, and intended maneuvers. This group was designed to demonstrate how enhanced communication could facilitate more efficient decision-making processes among vehicles, thereby reducing the likelihood of conflicts. The distinction between these groups was pivotal in assessing the advantages of V2X technology, as it allowed for direct comparisons of traffic outcomes under varying cooperative conditions. This experimental design aimed to provide a clear understanding of how V2X cooperation could enhance traffic flow and safety at unsignalized intersections, which is an intensive research focus [10][11].

C. Experimental Setup

The experimental setup was meticulously designed to ensure that each simulation run would yield reliable and meaningful data. Each simulation was conducted over a

duration of 180 seconds, a timeframe selected to capture a representative sample of vehicle interactions at the intersection. The number of vehicles in each simulation was systematically varied, ranging from 10 to 35, allowing for the examination of traffic density effects on performance metrics.

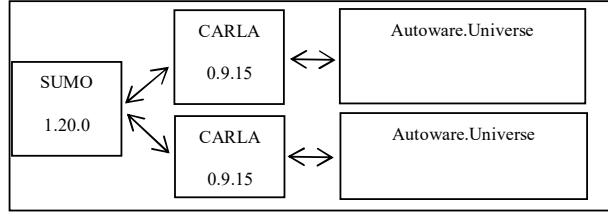


Figure 3. Diagram of One-SUMO-Two-CARLA

We have connected a SUMO server (version 1.20.0) to two CARLA clients (version 0.9.15), and each CARLA client is linked to an Autoware instance as shown in Fig. 3. In this way, we can control the vehicles in Autoware to create two autonomous vehicles in CARLA. By carefully setting up the scenarios, we can position these two autonomous vehicles at a T-junction simultaneously, thereby creating a scenario where autonomous vehicles encounter each other at the intersection.

The default configurations of Autoware and CARLA's built-in vehicle dynamics models were utilized to maintain consistency across simulations. These configurations included predefined parameters for vehicle acceleration, braking, and turning behaviors, ensuring that the vehicles behaved realistically during the experiments. Evaluation metrics were carefully chosen to assess traffic efficiency, including average speed, total travel time, and average travel time, all of which were collected using SUMO's statistical functions.

IV. EXPERIMENTAL ANALYSIS

What is important here is the fact that by using the proposed co-simulation platform, it is possible to conduct relatively detailed experimental analysis and quickly validate ideas without the need for extensive code development.

A. Comparison of Traffic Efficiency

The experimental results revealed notable differences in traffic efficiency between the two groups. In the absence of V2X cooperation, the frequency of deadlocks increased significantly as vehicle numbers rose.

In contrast, with V2X cooperation, traffic remained relatively smooth even with the increased vehicles, highlighting V2X technology's potential to enhance traffic efficiency through improved vehicle coordination.

B. Typical Scenario Analysis

To further illustrate the benefits of V2X cooperation, a series of typical scenarios were constructed using our joint simulation platform, each reflecting common situations encountered at unsignalized intersections. One particularly telling scenario involved four autonomous vehicles attempting to enter the intersection simultaneously, which resulted in a deadlock in the non-cooperation group. The vehicles, lacking communication, were unable to negotiate right-of-way effectively, leading to a longer average waiting time for each

vehicle involved. This situation starkly highlighted the limitations of relying solely on onboard sensing and decision-making in complex traffic environments.

V. CONCLUSIONS

This study successfully established a joint simulation platform utilizing SUMO, CARLA, and Autoware, enabling comprehensive analyses of multi-vehicle interactions at unsignalized intersections. The results confirm that V2X cooperation technology can effectively mitigate deadlock issues and enhance overall traffic efficiency. The joint simulation platform represents a robust tool for researching and optimizing autonomous driving systems, with potential applications in more complex traffic scenarios and algorithm validation. Future work may extend this platform to explore the performance of various autonomous driving algorithms in increasingly complex environments and investigate additional advanced technologies to further improve traffic efficiency and safety. In the future, by creating a comprehensive array of scenarios using the presented joint simulation platform, we can thoroughly evaluate the performance of autonomous driving algorithms in managing traffic conflicts, ultimately providing insights into their effectiveness in real-world applications.

REFERENCES

- [1] E. Yurtsever, J. Lambert, A. Carballo and K. Takeda, "A Survey of Autonomous Driving: Common Practices and Emerging Technologies," in *IEEE Access*, vol. 8, pp. 58443-58469, 2020.
- [2] K. G. Lim, C. H. Lee, R. K. Y. Chin, K. B. Yeo, and K. T. K. Teo, "SUMO enhancement for vehicular ad hoc network (VANET) simulation," in *Proc. IEEE 2nd Int. Conf. Autom. Control Intell. Syst.*, 2017, pp. 86-91.
- [3] D. Krajzewicz, J. Erdmann, M. Behrisch, and L. Bieker, "Recent Development and Applications of SUMO - Simulation of Urban Mobility," *Int. J. Adv. Syst. Meas.*, vol. 5, pp. 128-138, 2012.
- [4] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "CARLA: An Open Urban Driving Simulator," in *Proc. 1st Annu. Conf. Robot Learn.*, pp. 1-16, 2017.
- [5] S. Kato et al., "Autoware on board: Enabling autonomous vehicles with embedded systems," in *Proc. ACM/IEEE 9th Int. Conf. Cyber-Phys. Syst.*, 2018, pp. 287-296.
- [6] S. Bachuwar, A. Bulsara, H. Dossaji, A. Gopinath, and et al., "Integration of Autonomous Vehicle Frameworks for Software-in-the-Loop Testing," *SAE Int. J. Adv. Curr. Prac. Mobil.*, vol. 2, no. 5, pp. 2617-2622, 2020.
- [7] J. Zhou, Y. Zhang, S. Guo, and Y. Guo, "A Survey on Autonomous Driving System Simulators," in *2022 IEEE Int. Symp. Softw. Reliab. Eng. Workshops (ISSREW)*, Oct. 2022 - Nov. 3, 2022.
- [8] L. Wang, Y. Guo, H. Li, and Y. Liu, "Mixed Traffic Flow Characteristics Analysis Based on Simulation Approach," in *Proc. SPIE 13064, Seventh Int. Conf. Traffic Eng. Transp. Syst. (ICTETS 2023)*, 130641E, Feb. 20, 2024.
- [9] C. Y. Kuo and W. Liang, "Running Multiple Autoware-Powered Vehicles in Carla using Zenoh," Autoware, 2023. [Online]. Available: <https://autoware.org/running-multiple-autoware-powered-vehicles-in-carla-using-zenoh/>.
- [10] C. Jung, D. Lee, S. Lee, and D. H. Shim, "V2X-Communication-Aided Autonomous Driving: System Design and Experimental Validation," *Sensors*, vol. 20, no. 10, pp. 2903, 2020.
- [11] Q. Ge, G. Sartoretti, J. Duan, S. E. Li, Y. Yin and S. Zheng, "Distributed Model Predictive Control of Connected Multi-Vehicle Systems at Unsignalized Intersections," *2022 IEEE International Conference on Unmanned Systems (ICUS)*, Guangzhou, China, 2022, pp. 1466-1472.